# MSG No. 189.222 $P\bar{6}2m1'$ [ Type II, hexagonal ]

Table 1: Wyckoff site: `1a`, site symmetry: `-62m1'`

| No. | position | mapping |
|---|---|---|
| 1 | [0, 0, 0] | [1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24] |

Table 2: Wyckoff site: `1b`, site symmetry: `-62m1'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, 0, \frac{1}{2}\right]$ | [1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24] |

Table 3: Wyckoff site: `2c`, site symmetry: `-6..1'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3}, \frac{2}{3}, 0\right]$ | [1,2,3,7,8,9,13,14,15,19,20,21] |
| 2 | $\left[\frac{2}{3}, \frac{1}{3}, 0\right]$ | [4,5,6,10,11,12,16,17,18,22,23,24] |

Table 4: Wyckoff site: `2d`, site symmetry: `-6..1'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$ | [1,2,3,7,8,9,13,14,15,19,20,21] |
| 2 | $\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$ | [4,5,6,10,11,12,16,17,18,22,23,24] |

Table 5: Wyckoff site: `2e`, site symmetry: `3.m1'`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, z]$ | [1,2,3,10,11,12,13,14,15,22,23,24] |
| 2 | $[0, 0, -z]$ | [4,5,6,7,8,9,16,17,18,19,20,21] |

Table 6: Wyckoff site: `3f`, site symmetry: `m2m1'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x, 0, 0]$ | [1,4,8,11,13,16,20,23] |
| 2 | $[0, x, 0]$ | [2,5,9,12,14,17,21,24] |
| 3 | $[-x, -x, 0]$ | [3,6,7,10,15,18,19,22] |

Table 7: Wyckoff site: `3g`, site symmetry: `m2m1'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x, 0, \frac{1}{2}\right]$ | [1,4,8,11,13,16,20,23] |
| 2 | $\left[0, x, \frac{1}{2}\right]$ | [2,5,9,12,14,17,21,24] |

| | | |
|---|---|---|
| 3 | $\left[-x,\, -x,\, \frac{1}{2}\right]$ | [3,6,7,10,15,18,19,22] |

Table 8: Wyckoff site: 4h, site symmetry: 3..1'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{3},\, \frac{2}{3},\, z\right]$ | [1,2,3,13,14,15] |
| 2 | $\left[\frac{2}{3},\, \frac{1}{3},\, -z\right]$ | [4,5,6,16,17,18] |
| 3 | $\left[\frac{1}{3},\, \frac{2}{3},\, -z\right]$ | [7,8,9,19,20,21] |
| 4 | $\left[\frac{2}{3},\, \frac{1}{3},\, z\right]$ | [10,11,12,22,23,24] |

Table 9: Wyckoff site: 6i, site symmetry: ..m1'

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, 0,\, z]$ | [1,11,13,23] |
| 2 | $[0,\, x,\, z]$ | [2,12,14,24] |
| 3 | $[-x,\, -x,\, z]$ | [3,10,15,22] |
| 4 | $[x,\, 0,\, -z]$ | [4,8,16,20] |
| 5 | $[0,\, x,\, -z]$ | [5,9,17,21] |
| 6 | $[-x,\, -x,\, -z]$ | [6,7,18,19] |

Table 10: Wyckoff site: 6j, site symmetry: m..1'

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, y,\, 0]$ | [1,8,13,20] |
| 2 | $[-y,\, x-y,\, 0]$ | [2,9,14,21] |
| 3 | $[-x+y,\, -x,\, 0]$ | [3,7,15,19] |
| 4 | $[x-y,\, -y,\, 0]$ | [4,11,16,23] |
| 5 | $[y,\, x,\, 0]$ | [5,12,17,24] |
| 6 | $[-x,\, -x+y,\, 0]$ | [6,10,18,22] |

Table 11: Wyckoff site: 6k, site symmetry: m..1'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\, y,\, \frac{1}{2}\right]$ | [1,8,13,20] |
| 2 | $\left[-y,\, x-y,\, \frac{1}{2}\right]$ | [2,9,14,21] |
| 3 | $\left[-x+y,\, -x,\, \frac{1}{2}\right]$ | [3,7,15,19] |
| 4 | $\left[x-y,\, -y,\, \frac{1}{2}\right]$ | [4,11,16,23] |
| 5 | $\left[y,\, x,\, \frac{1}{2}\right]$ | [5,12,17,24] |
| 6 | $\left[-x,\, -x+y,\, \frac{1}{2}\right]$ | [6,10,18,22] |

Table 12: Wyckoff site: 12l, site symmetry: 11'

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | [1,13] |
| 2 | $[-y,\ x-y,\ z]$ | [2,14] |
| 3 | $[-x+y,\ -x,\ z]$ | [3,15] |
| 4 | $[x-y,\ -y,\ -z]$ | [4,16] |
| 5 | $[y,\ x,\ -z]$ | [5,17] |
| 6 | $[-x,\ -x+y,\ -z]$ | [6,18] |
| 7 | $[-x+y,\ -x,\ -z]$ | [7,19] |
| 8 | $[x,\ y,\ -z]$ | [8,20] |
| 9 | $[-y,\ x-y,\ -z]$ | [9,21] |
| 10 | $[-x,\ -x+y,\ z]$ | [10,22] |
| 11 | $[x-y,\ -y,\ z]$ | [11,23] |
| 12 | $[y,\ x,\ z]$ | [12,24] |